

# Executive Summary

This audit report was prepared by Quantstamp, the leader in blockchain security.

|               |                                                                                                                                                                                          |                                  |        |                                |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|--------|--------------------------------|
| Type          | ERC-6900 Module                                                                                                                                                                          | Documentation quality            | Medium | <div><div></div></div>         |
| Timeline      | 2025-09-17 through 2025-09-29                                                                                                                                                            | Test quality                     | High   | <div><div></div></div>         |
| Language      | Solidity                                                                                                                                                                                 | Total Findings                   | 1      | <div><div></div>Fixed: 1</div> |
| Methods       | Architecture Review, Unit Testing, Functional Testing, Computer-Aided Verification, Manual Review                                                                                        | High severity findings ⓘ         | 0      |                                |
| Specification | Private documentation provided                                                                                                                                                           | Medium severity findings ⓘ       | 0      |                                |
| Source Code   | <ul style="list-style-type: none"><li><a href="#">#github.com/exactly/exa diff from 1a27efdfb95e76e711251670d4fb5fa66a90ce2d to 16efc1be67e03dcd407ad3430150456dac965434</a> ↗</li></ul> | Low severity findings ⓘ          | 0      |                                |
| Auditors      | <ul style="list-style-type: none"><li>Ruben Koch Senior Auditing Engineer</li><li>Hytham Farah Auditing Engineer</li><li>Nikita Belenkov Senior Auditing Engineer</li></ul>              | Undetermined severity findings ⓘ | 0      |                                |
|               |                                                                                                                                                                                          | Informational findings ⓘ         | 1      | <div><div></div>Fixed: 1</div> |

# Summary of Findings

In this audit, we have reviewed a diff based on the previously audited Exa plugin module between the `1a27efdfb95e76e711251670d4fb5fa66a90ce2d` and `16efc1be67e03dcd407ad3430150456dac965434` commits. This change has added the ability to execute multiple proposals in one transaction, cross-repay with non-USDC markets, and other minor fixes.

Only one issue has been identified that should be fixed before deployment. The test suite has been updated to reflect the changes introduced in the diff.

### Fix-Review Update 2025-10-30:

The identified finding, alongside a few minor miscellaneous changes, have been reviewed up until commit `f31ac5993466eab0637550f2a3f5d438b205bdc8`. Tests have been added to verify the fix.

| ID    | DESCRIPTION                                                                   | SEVERITY          | STATUS |
|-------|-------------------------------------------------------------------------------|-------------------|--------|
| EXA-1 | <code>marketOut</code> in Cross Repay-Flow Should Always Be Registered Market | • Informational ⓘ | Fixed  |

# Assessment Breakdown

Quantstamp's objective was to evaluate the repository for security-related issues, code quality, and adherence to specification and best practices.



### Disclaimer

Only features that are contained within the repositories at the commit hashes specified on the front page of the report are within the scope of the audit and fix review. All features added in future revisions of the code are excluded from consideration in this report.

Possible issues we looked for included (but are not limited to):

- Transaction-ordering dependence
- Timestamp dependence
- Mishandled exceptions and call stack limits
- Unsafe external calls
- Integer overflow / underflow
- Number rounding errors
- Reentrancy and cross-function vulnerabilities
- Denial of service / logical oversights
- Access control
- Centralization of power
- Business logic contradicting the specification
- Code clones, functionality duplication
- Gas usage
- Arbitrary token minting

Methodology

1. Code review that includes the following
  1. Review of the specifications, sources, and instructions provided to Quantstamp to make sure we understand the size, scope, and functionality of the smart contract.
  2. Manual review of code, which is the process of reading source code line-by-line in an attempt to identify potential vulnerabilities.
  3. Comparison to specification, which is the process of checking whether the code does what the specifications, sources, and instructions provided to Quantstamp describe.
2. Testing and automated analysis that includes the following:
  1. Test coverage analysis, which is the process of determining whether the test cases are actually covering the code and how much code is exercised when we run those test cases.
  2. Symbolic execution, which is analyzing a program to determine what inputs cause each part of a program to execute.
3. Best practices review, which is a review of the smart contracts to improve efficiency, effectiveness, clarity, maintainability, security, and control based on the established industry and academic practices, recommendations, and research.
4. Specific, itemized, and actionable recommendations to help you take steps to secure your smart contracts.

Scope

Scope is the diff between the following commits: 1a27efdfb95e76e711251670d4fb5fa66a90ce2d and 16efc1be67e03dcd407ad3430150456dac965434 , which included the following files in the scope:

1. contracts/src/ExaPlugin.sol
2. contracts/src/ExaPluginExtension.sol
3. contracts/src/IExaAccount.sol
4. contracts/src/ProposalManager.sol

Files Included

Repo: https://github.com/exactly/exa

Included Paths: contracts/src/ExaPlugin.sol , contracts/src/ExaPluginExtension.sol , contracts/src/IExaAccount.sol , contracts/src/ProposalManager.sol

Findings

EXA-1

marketOut in Cross Repay-Flow Should Always Be Registered Market

• Informational ⓘ Fixed

*i*

Update

Marked as "Fixed" by the client.

Addressed in: c132d697a52b1e796eb625df0fd7696dd93dc8e3 .

File(s) affected: contracts/src/ProposalManager.sol

**Description:** With the updated code, cross-repay is now possible into arbitrary markets instead of just the USDC markets. This is done with a new marketOut field in the CrossRepayCallbackData struct. However, that marketOut field is not undergoing a \_checkMarket() check in the propose() flow in the ProposalManager contract. While no clear attack vectors have been identified, we recommend performing such a check to make sure that the plugin operates in the restrictions of the registered markets as defined by the AUDITOR instance, especially since the underlying marketOut.asset() is used in an external call via flash loans.

**Recommendation:** In the `propose()` function, in case the `proposalType` is `ProposalType.CROSS_REPAY_AT_MATURITY`, decode the `CrossRepayData` from the `proposal.data` and assure that `_isMarket(marketOut)` is given.

# Definitions

- **High severity** – High-severity issues usually put a large number of users' sensitive information at risk, or are reasonably likely to lead to catastrophic impact for client's reputation or serious financial implications for client and users.
- **Medium severity** – Medium-severity issues tend to put a subset of users' sensitive information at risk, would be detrimental for the client's reputation if exploited, or are reasonably likely to lead to moderate financial impact.
- **Low severity** – The risk is relatively small and could not be exploited on a recurring basis, or is a risk that the client has indicated is low impact in view of the client's business circumstances.
- **Informational** – The issue does not pose an immediate risk, but is relevant to security best practices or Defence in Depth.
- **Undetermined** – The impact of the issue is uncertain.
- **Fixed** – Adjusted program implementation, requirements or constraints to eliminate the risk.
- **Mitigated** – Implemented actions to minimize the impact or likelihood of the risk.
- **Acknowledged** – The issue remains in the code but is a result of an intentional business or design decision. As such, it is supposed to be addressed outside the programmatic means, such as: 1) comments, documentation, README, FAQ; 2) business processes; 3) analyses showing that the issue shall have no negative consequences in practice (e.g., gas analysis, deployment settings).

# Appendix

## File Signatures

The following are the SHA-256 hashes of the reviewed files. A file with a different SHA-256 hash has been modified, intentionally or otherwise, after the security review. You are cautioned that a different SHA-256 hash could be (but is not necessarily) an indication of a changed condition or potential vulnerability that was not within the scope of the review.

## Files

- Repo: `https://github.com/exactly/exa`
- `c1a...012 ./contracts/src/ExaPlugin.sol`
  - `fcf...e00 ./contracts/src/ExaPluginExtension.sol`
  - `0bc...4b2 ./contracts/src/IExaAccount.sol`
  - `2a4...a8d ./contracts/src/ProposalManager.sol`

# Test Suite Results

The test suite consists of 220 tests, of which all pass.

**Fix-Review Update:** A few tests have been added to test the now less restricted cross-repay functionality.

```
Ran 8 tests for test/IssuerChecker.t.sol:IssuerCheckerTest
[PASS] test_setIssuer_emits_IssuerSet() (gas: 70595)
[PASS] test_setIssuer_reverts_whenNotAdmin() (gas: 37043)
[PASS] test_setIssuer_reverts_whenZeroAddress() (gas: 35466)
[PASS] test_setOperationExpiry_emits_OperationExpirySet() (gas: 46742)
[PASS] test_setOperationExpiry_reverts_whenNotAdmin() (gas: 38913)
[PASS] test_setOperationExpiry_reverts_whenZeroValue() (gas: 37400)
[PASS] test_setPrevIssuerWindow_emits_PrevIssuerWindowSet() (gas: 46777)
[PASS] test_setPrevIssuerWindow_reverts_whenNotAdmin() (gas: 38890)
Suite result: ok. 8 passed; 0 failed; 0 skipped; finished in 211.66ms (353.67ms CPU time)

Ran 2 tests for test/MockSwapper.t.sol:MockSwapperTest
[PASS] test_swapExactAmountIn_swaps() (gas: 247315)
[PASS] test_swapExactAmountOut_swaps() (gas: 247311)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 219.90ms (2.67ms CPU time)

Ran 9 tests for test/ExaPreviewer.t.sol:ExaPreviewerTest
[PASS] test_assets_returnsAssets() (gas: 67641)
```

```
[PASS] test_collectCollateral_collects() (gas: 1140814)
[PASS] test_collectInstallments_collects() (gas: 1152032)
[PASS] test_collect_reverts_whenProposalsHaveTooMuchDebt() (gas: 2067840)
[PASS] test_collect_reverts_whenProposalsHaveTooMuchRollDebt() (gas: 873114)
[PASS] test_collect_reverts_whenProposalsLeaveNoLiquidity() (gas: 1376581)
[PASS] test_markets_returnsMarkets() (gas: 132537)
[PASS] test_pendingProposals_returnsPendingProposals() (gas: 1870986)
[PASS] test_utilizations_returns() (gas: 135053)
Suite result: ok. 9 passed; 0 failed; 0 skipped; finished in 294.18ms (181.04ms CPU time)

Ran 9 tests for test/Refunder.t.sol:RefunderTest
[PASS] test_refund_refunds() (gas: 246369)
[PASS] test_refund_reverts_whenExpired() (gas: 73677)
[PASS] test_refund_reverts_whenNotKeeper() (gas: 56679)
[PASS] test_refund_reverts_whenReplay() (gas: 285456)
[PASS] test_refund_reverts_whenTimelocked() (gas: 63242)
[PASS] test_refund_reverts_whenWrongSignature() (gas: 95260)
[PASS] test_refund_toleratesTimeDrift() (gas: 246500)
[PASS] test_withdraw_reverts_whenNotAdmin() (gas: 35768)
[PASS] test_withdraw_withdraws() (gas: 68494)
Suite result: ok. 9 passed; 0 failed; 0 skipped; finished in 295.04ms (323.72ms CPU time)

Ran 2 tests for test/ExaAccountFactory.t.sol:ExaAccountFactoryTest
[PASS] testFuzz_createAccount_EOAOwners(uint256,address[63]) (runs: 256, μ: 4211337, ~: 4224752)
[PASS] test_deploy_deploysToSameAddress() (gas: 13327133)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 1.16s (1.68s CPU time)

Ran 190 tests for test/ExaPlugin.t.sol:ExaPluginTest
[PASS] testFork_claimAndVestEscrowedEXA_claimsAndVests() (gas: 37240819)
[PASS] testFork_collectCollateral_collects() (gas: 31456303)
[PASS] testFork_crossRepay_repays() (gas: 33217903)
[PASS] testFork_stakeEXA_stakes() (gas: 32037473)
[PASS] testFork_swap_swaps() (gas: 27521916)
[PASS] test_allowPlugin_emitsPluginAllowed() (gas: 58441)
[PASS] test_allowPlugin_reverts_whenAddressZero() (gas: 32411)
[PASS] test_allowPlugin_reverts_whenNotAdmin() (gas: 33988)
[PASS] test_allowPlugin_sets_whenAdmin() (gas: 57860)
[PASS] test_allowTarget_emitsTargetAllowed() (gas: 82849)
[PASS] test_allowTarget_reverts_whenAddressZero() (gas: 32349)
[PASS] test_allowTarget_reverts_whenNotAdmin() (gas: 38387)
[PASS] test_allowTarget_sets_whenAdmin() (gas: 54647)
[PASS] test_approveMarket_reverts_whenNotAllowlisted() (gas: 95078)
[PASS] test_auditorCalls_passes() (gas: 117651)
[PASS] test_borrowAtMaturity_borrows() (gas: 1132711)
[PASS] test_borrowAtMaturity_borrowsEXA() (gas: 1049563)
[PASS] test_borrowAtMaturity_incrementsNonce() (gas: 1149682)
[PASS] test_borrowAtMaturity_reverts_whenProposalTypeIsWithdraw() (gas: 637350)
[PASS] test_borrowAtMaturity_reverts_whenTimelocked() (gas: 691634)
[PASS] test_borrowAtMaturity_reverts_withNoProposal_whenDataDiffers() (gas: 695879)
[PASS] test_borrowAtMaturity_reverts_withNoProposal_whenReceiverNotCollector() (gas: 388620)
[PASS] test_borrowToCollector_borrows() (gas: 695477)
[PASS] test_borrow_reverts_withUnauthorized_whenReceiverNotCollector() (gas: 383663)
[PASS] test_collectCollateral_collects() (gas: 820997)
[PASS] test_collectCollateral_reverts_withDisagreement() (gas: 822527)
[PASS] test_collectCredit_collects() (gas: 842011)
[PASS] test_collectCredit_collects_whenHealthFactorHigherThanOne() (gas: 735188)
[PASS] test_collectCredit_collects_whenProposalCausesInsufficientLiquidity() (gas: 996322)
[PASS] test_collectCredit_collects_whenProposalLeavesHealthFactorLowerThanOne() (gas: 990897)
[PASS] test_collectCredit_collects_withEnoughSlippage() (gas: 733726)
[PASS] test_collectCredit_collects_withPrevIssuerSignature() (gas: 920031)
[PASS] test_collectCredit_passes_whenProposalLeavesEnoughLiquidity() (gas: 1000928)
[PASS] test_collectCredit_reverts_asNotKeeper() (gas: 339723)
[PASS] test_collectCredit_reverts_whenDisagreement() (gas: 536742)
[PASS] test_collectCredit_reverts_whenExpired() (gas: 343374)
[PASS] test_collectCredit_reverts_whenPrevSignatureNotValidAnymore() (gas: 465094)
[PASS] test_collectCredit_reverts_whenReplay() (gas: 778896)
[PASS] test_collectCredit_reverts_whenTimelocked() (gas: 332163)
[PASS] test_collectCredit_toleratesTimeDrift() (gas: 737125)
[PASS] test_collectDebit_collects() (gas: 592139)
[PASS] test_collectDebit_collects_whenProposalCausesInsufficientLiquidity() (gas: 851555)
[PASS] test_collectDebit_collects_whenProposalLeavesEnoughLiquidity() (gas: 852995)
[PASS] test_collectDebit_reverts_asNotKeeper() (gas: 339534)
```



[PASS] test\_collectDebit\_reverts\_whenExpired() (gas: 343081)  
[PASS] test\_collectDebit\_reverts\_whenReplay() (gas: 633677)  
[PASS] test\_collectDebit\_reverts\_whenTimelocked() (gas: 331852)  
[PASS] test\_collectDebit\_toleratesTimeDrift() (gas: 592247)  
[PASS] test\_collectInstallments\_collects() (gas: 1229370)  
[PASS] test\_collectInstallments\_revertsWhenNoSlippage() (gas: 1172648)  
[PASS] test\_collectInstallments\_reverts\_asNotKeeper() (gas: 340596)  
[PASS] test\_collectInstallments\_reverts\_whenExpired() (gas: 345488)  
[PASS] test\_collectInstallments\_reverts\_whenReplay() (gas: 976617)  
[PASS] test\_collectInstallments\_reverts\_whenTimelocked() (gas: 334332)  
[PASS] test\_collectInstallments\_toleratesTimeDrift() (gas: 1098471)  
[PASS] test\_collect\_collects\_whenProposalsLeaveNoLiquidity() (gas: 1719084)  
[PASS] test\_collect\_collects\_whenTooMuchProposedDebt() (gas: 1800029)  
[PASS] test\_crossRepay\_avoidsFrozenDeposit() (gas: 886722)  
[PASS] test\_crossRepay\_avoidsZeroSharesDeposit() (gas: 1307325)  
[PASS] test\_crossRepay\_consumesProposal() (gas: 2094054)  
[PASS] test\_crossRepay\_repays() (gas: 2105330)  
[PASS] test\_crossRepay\_repaysEXA() (gas: 2683750)  
[PASS] test\_crossRepay\_repays\_whenKeeper() (gas: 2104316)  
[PASS] test\_crossRepay\_reverts\_whenDisagreement() (gas: 2171495)  
[PASS] test\_crossRepay\_reverts\_whenNotKeeper() (gas: 1382716)  
[PASS] test\_debtManagerCalls\_revert\_withUnauthorized\_whenSelectorNotRollFixed() (gas: 74399)  
[PASS] test\_exaPluginConstructor\_reverts\_whenSwapperIsZeroAddress() (gas: 1705803)  
[PASS] test\_executeBatch\_checksEveryCall() (gas: 242408)  
[PASS] test\_executeBatch\_reverts\_whenUnauthorizedCallsNested() (gas: 152386)  
[PASS] test\_executeBatch\_reverts\_whenWithdrawingMultipleWithOneProposal() (gas: 718925)  
[PASS] test\_executeBatch\_supportsMultipleProposals() (gas: 1589395)  
[PASS] test\_executeProposal\_reverts\_whenInvalidNonce() (gas: 298478)  
[PASS] test\_executeProposals\_executes\_whenAccount() (gas: 1158072)  
[PASS] test\_executeProposals\_executes\_whenKeeper() (gas: 1158524)  
[PASS] test\_executeProposals\_reverts\_whenNoProposal() (gas: 1156361)  
[PASS] test\_executeProposals\_reverts\_whenNotKeeperOrAccount() (gas: 483467)  
[PASS] test\_executeProposals\_reverts\_whenNotNext() (gas: 505372)  
[PASS] test\_exitMarket\_reverts() (gas: 366793)  
[PASS] test\_forceUninstall\_reverts\_withUnauthorized\_withoutExecute() (gas: 72040)  
[PASS] test\_install\_installs\_whenNotRelatedPlugin() (gas: 1481657)  
[PASS] test\_install\_reverts\_whenAlreadyInstalled() (gas: 6737809)  
[PASS] test\_installmentsRouterCalls\_revert\_withUnauthorized\_whenSelectorNotBorrow() (gas: 75722)  
[PASS] test\_issuerChecker\_emits\_collected() (gas: 834799)  
[PASS] test\_issuerChecker\_emits\_refunded() (gas: 355001)  
[PASS] test\_marketWithdraw\_transfersAsset\_asOwner() (gas: 892668)  
[PASS] test\_pluginMetadata() (gas: 12946)  
[PASS] test\_poke() (gas: 292926)  
[PASS] test\_pokeETH\_deposits() (gas: 349100)  
[PASS] test\_poke\_reverts\_withNoBalance() (gas: 126728)  
[PASS] test\_poke\_reverts\_withNotMarket() (gas: 63216)  
[PASS] test\_postExecutionHook\_reverts\_withNotImplemented() (gas: 30761)  
[PASS] test\_preExecutionHook\_reverts\_withNotImplemented() (gas: 33544)  
[PASS] test\_proposeRepay\_reverts\_whenWrongType() (gas: 53482)  
[PASS] test\_proposeSwap\_emitsSwapProposed() (gas: 399179)  
[PASS] test\_proposeWithdraw\_reverts\_whenKeeper() (gas: 331337)  
[PASS] test\_proposeZeroAmount\_reverts\_withZeroAmount() (gas: 40739)  
[PASS] test\_propose\_emitsProposed() (gas: 281777)  
[PASS] test\_propose\_incrementsQueueNonces() (gas: 178225)  
[PASS] test\_propose\_reverts\_whenNoneType() (gas: 58367)  
[PASS] test\_propose\_reverts\_whenNotMarket() (gas: 896654)  
[PASS] test\_propose\_reverts\_whenNotProposer() (gas: 39528)  
[PASS] test\_redeem\_reverts\_whenNoProposalAndReceiverIsProposer() (gas: 436252)  
[PASS] test\_redeem\_reverts\_whenProposalTypeIsWithdraw() (gas: 640613)  
[PASS] test\_redeem\_withdraws() (gas: 900131)  
[PASS] test\_redeem\_withdraws\_whenExecuteProposal() (gas: 944872)  
[PASS] test\_refund\_refunds() (gas: 365306)  
[PASS] test\_refund\_refunds\_whenCollectedAtSameTime() (gas: 906523)  
[PASS] test\_repay\_consumesProposal() (gas: 1305432)  
[PASS] test\_repay\_partiallyRepays() (gas: 1355209)  
[PASS] test\_repay\_partiallyRepays\_whenKeeper() (gas: 1329349)  
[PASS] test\_repay\_repaysX() (gas: 1329867)  
[PASS] test\_repay\_repays\_anyMarket() (gas: 1596109)  
[PASS] test\_repay\_repays\_whenKeeper() (gas: 1293330)  
[PASS] test\_rollDebt\_rollsX() (gas: 1505521)  
[PASS] test\_rollDebt\_rolls\_anyMarket() (gas: 1788173)  
[PASS] test\_rollDebt\_rolls\_asKeeper() (gas: 1507621)

```
[PASS] test_rollFixed_reverts__withNoProposal_proposalTypeDiffers() (gas: 340407)
[PASS] test_rollFixed_reverts_whenTimelocked() (gas: 342577)
[PASS] test_rollFixed_reverts_withNoProposal_whenRollDataDiffers() (gas: 421646)
[PASS] test_runtimeValidationFunction_reverts_withNotImplemented() (gas: 9525)
[PASS] test_setCollector_emitsCollectorSet() (gas: 40522)
[PASS] test_setCollector_reverts_whenAddressZero() (gas: 32187)
[PASS] test_setCollector_reverts_whenNotAdmin() (gas: 33740)
[PASS] test_setCollector_sets_whenAdmin() (gas: 39824)
[PASS] test_setDelay_emitsDelaySet() (gas: 39425)
[PASS] test_setDelay_reverts_whenHigherThanOneHours() (gas: 32131)
[PASS] test_setDelay_reverts_whenNotAdmin() (gas: 38123)
[PASS] test_setDelay_reverts_whenZero() (gas: 32116)
[PASS] test_setDelay_sets_whenAdmin() (gas: 39048)
[PASS] test_setFlashLoaner_emitsFlashLoanerSet() (gas: 40486)
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[PASS] test_setProposalManager_emitsProposalManagerSet() (gas: 40455)
[PASS] test_setProposalManager_reverts_whenAddressZero() (gas: 32144)
[PASS] test_setProposalManager_reverts_whenNotAdmin() (gas: 33699)
[PASS] test_setProposalManager_sets_whenAdmin() (gas: 39759)
[PASS] test_setProposalNonce_emitsEvent() (gas: 324314)
[PASS] test_setProposalNonce_sets() (gas: 1174633)
[PASS] test_setSwapper_emitsSwapperSet() (gas: 40456)
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[PASS] test_setSwapper_reverts_whenNotAdmin() (gas: 33655)
[PASS] test_setSwapper_sets_whenAdmin() (gas: 39759)
[PASS] test_skippingProposal_reverts_withNoProposal() (gas: 1993841)
[PASS] test_swap_reverts_whenAssetInIsMarket() (gas: 373424)
[PASS] test_swap_reverts_whenCallerIsNotSelf() (gas: 52147)
[PASS] test_swap_reverts_withDisagreement() (gas: 335053)
[PASS] test_swap_swaps() (gas: 309002)
[PASS] test_targetNotAllowed_reverts_withUnauthorized() (gas: 1095064)
[PASS] test_transferShares_reverts_whenNoRightProposal() (gas: 633945)
[PASS] test_transferShares_transfers_whenRedeemProposalPresent() (gas: 783215)
[PASS] test_uninstallAndInstall_installs_whenPluginIsAllowed() (gas: 677713)
[PASS] test_uninstall_reverts_whenExecuteBatchInsideExecute() (gas: 106868)
[PASS] test_uninstall_reverts_whenInsideExecuteOfExecuteBatch() (gas: 148729)
[PASS] test_uninstall_reverts_whenPendingProposals() (gas: 559771)
[PASS] test_uninstall_reverts_whitUnauthorized_whenNextCallIsNotInstall() (gas: 74460)
[PASS] test_uninstall_reverts_withUnauthorized_whenExecute() (gas: 63430)
[PASS] test_uninstall_reverts_withUnauthorized_whenInsideBatch() (gas: 66481)
[PASS] test_uninstall_reverts_withUnauthorized_whenNotAllowedPlugin() (gas: 153027)
[PASS] test_uninstall_reverts_withUnauthorized_withoutExecute() (gas: 70507)
[PASS] test_uninstall_uninstalls_whenBatchedWithAnotherPlugin() (gas: 931149)
[PASS] test_uninstall_uninstalls_whenExecuteWithAnotherPlugin() (gas: 926280)
[PASS] test_uninstall_uninstalls_whenItsAnotherPlugin() (gas: 900001)
[PASS] test_uninstall_uninstalls_whenWrongProposalManager() (gas: 660633)
[PASS] test_userOpValidationFunction_reverts_withBadPlugin() (gas: 1805611)
[PASS] test_withdrawAtMaturity_withdraws() (gas: 465435)
[PASS] test_withdrawETH_asOwner_consumesProposal() (gas: 899150)
[PASS] test_withdrawETH_consumesProposal() (gas: 899152)
[PASS] test_withdrawProposed_emits_proposalNonceSet() (gas: 890861)
[PASS] test_withdrawProposed_incrementsNonce() (gas: 889829)
[PASS] test_withdraw_reverts_whenNoProposal() (gas: 441782)
[PASS] test_withdraw_reverts_whenNoProposalAndReceiverIsProposer() (gas: 438437)
[PASS] test_withdraw_reverts_whenNoProposalKeeper() (gas: 399133)
[PASS] test_withdraw_reverts_whenNotKeeper() (gas: 337250)
[PASS] test_withdraw_reverts_whenProposalTypeIsBorrowAtMaturity() (gas: 643011)
[PASS] test_withdraw_reverts_whenProposalTypeIsRedeem() (gas: 643043)
[PASS] test_withdraw_reverts_whenReceiverIsContractAndMarketNotWETH() (gas: 695304)
[PASS] test_withdraw_reverts_whenTimelocked() (gas: 401532)
[PASS] test_withdraw_reverts_whenTimelockedETH() (gas: 646604)
[PASS] test_withdraw_reverts_whenTimelockedKeeper() (gas: 345166)
[PASS] test_withdraw_reverts_whenWrongAmount() (gas: 403187)
[PASS] test_withdraw_reverts_whenWrongMarket() (gas: 401372)
[PASS] test_withdraw_reverts_whenWrongReceiver() (gas: 405377)
[PASS] test_withdraw_swapsProposed() (gas: 1669889)
[PASS] test_withdraw_swapsWETH() (gas: 1687036)
```

```
[PASS] test_withdraw_transfersETH() (gas: 906337)
[PASS] test_withdraw_withdrawsProposed() (gas: 865548)
Suite result: ok. 190 passed; 0 failed; 0 skipped; finished in 8.64s (8.63s CPU time)

Ran 6 test suites in 9.00s (10.83s CPU time): 220 tests passed, 0 failed, 0 skipped (220 total tests)
```

# Code Coverage

Branch coverage across the contracts is very good.

| File                       | % Lines              | % Statements         | % Branches         | % Funcs            |
|----------------------------|----------------------|----------------------|--------------------|--------------------|
| src/ExaPlugin.sol          | 99.63%<br>(266/267)  | 99.43%<br>(349/351)  | 98.04%<br>(50/51)  | 100.00%<br>(57/57) |
| src/ExaPluginExtension.sol | 100.00%<br>(48/48)   | 100.00%<br>(63/63)   | 88.89% (8/9)       | 100.00% (8/8)      |
| src/ProposalManager.sol    | 100.00%<br>(149/149) | 100.00%<br>(231/231) | 100.00%<br>(49/49) | 100.00%<br>(18/18) |

# Changelog

- 2025-09-30 - Initial report
- 2025-10-30 - Final report

# About Quantstamp

Quantstamp is a global leader in blockchain security. Founded in 2017, Quantstamp’s mission is to securely onboard the next billion users to Web3 through its best-in-class Web3 security products and services.

Quantstamp’s team consists of cybersecurity experts hailing from globally recognized organizations including Microsoft, AWS, BMW, Meta, and the Ethereum Foundation. Quantstamp engineers hold PhDs or advanced computer science degrees, with decades of combined experience in formal verification, static analysis, blockchain audits, penetration testing, and original leading-edge research.

To date, Quantstamp has performed more than 500 audits and secured over \$200 billion in digital asset risk from hackers. Quantstamp has worked with a diverse range of customers, including startups, category leaders and financial institutions. Brands that Quantstamp has worked with include Ethereum 2.0, Binance, Visa, PayPal, Polygon, Avalanche, Curve, Solana, Compound, Lido, MakerDAO, Arbitrum, OpenSea and the World Economic Forum.

Quantstamp’s collaborations and partnerships showcase our commitment to world-class research, development and security. We're honored to work with some of the top names in the industry and proud to secure the future of web3.

Notable Collaborations & Customers:

- Blockchains: Ethereum 2.0, Near, Flow, Avalanche, Solana, Cardano, Binance Smart Chain, Hedera Hashgraph, Tezos
- DeFi: Curve, Compound, Maker, Lido, Polygon, Arbitrum, SushiSwap
- NFT: OpenSea, Parallel, Dapper Labs, Decentraland, Sandbox, Axie Infinity, Illuvium, NBA Top Shot, Zora
- Academic institutions: National University of Singapore, MIT

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